SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	2024-AI&ML	**Date:**	17.09.2026

Code of Conduct

I __D.Sravanthi(192425315)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _D.Sravanthi________

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. **Given the following paragraph:**
   "John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

ANSWER:

Given Paragraph

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

a) Identify All Referring Expressions and Possible Antecedents

A **referring expression** is a word or phrase that refers to an entity mentioned earlier in the text.

Referring Expression	Type	Possible Antecedents
He	Pronoun	John, Mary
She	Pronoun	John, Mary
it	Pronoun	ball, park

Referring Expression	Type	Possible Antecedents
him	Pronoun	John, Mary, dog
they	Plural pronoun	John, Mary, dog
all	Group reference	John, Mary, dog

Explanation

- **"He"** refers to **John**.
- **"She"** refers to **Mary**.
- **"it"** refers to the **ball**.
- **"him"** refers to **John**.
- **"they"** refers to **John, Mary, and the dog**.
- **"all"** refers to the complete group of **John, Mary, and the dog**.

b) Applying the Given Constraints

1. Gender and Number Agreement

Gender and number agreement eliminates candidates that do not match the pronoun.

- **He** → **John** because John is male and singular.
- **She** → **Mary** because Mary is female and singular.
- **it** → **ball** because the ball is singular and inanimate.
- **him** → **John** because John is male and singular.
- **they** → **John + Mary + dog** because "they" is plural.

Therefore, gender and number agreement helps remove incorrect candidates.

2. Recency

The **recency constraint** prefers a recent and compatible antecedent.

For example, in:

"She wanted to play with it."

The word **"it"** could potentially refer to different objects, but **ball** is the most recent suitable object and is semantically related to playing.

Similarly:

“The dog chased him excitedly.”

The system considers recently mentioned entities, but gender agreement and semantic compatibility help select **John**.

3. Semantic Compatibility

Semantic compatibility checks whether the selected antecedent makes sense in the sentence.

- A person can **bring a ball**.
- Mary can **play with a ball**.
- A dog can **chase John**.
- John, Mary, and the dog can **go home together**.

Therefore:

- **it → ball**
- **him → John**
- **they → John + Mary + dog**

4. Coherence

Coherence ensures that the same entity is consistently referred to throughout the paragraph.

The main entity chains are:

- **John → He → him**
- **Mary → She**
- **Ball → it**
- **John + Mary + dog → they → all**

These chains maintain consistency throughout the paragraph.

c) Constraint Graph / Table

Constraint Table

Variable	Initial Domain	Gender/Number	Recency	Semantic Compatibility	Final Result
He	John, Mary	Mary eliminated	John preferred	John brought a ball	**John**
She	John, Mary	John eliminated	Mary preferred	Mary wants to play	**Mary**
it	ball, park	Both possible	Ball preferred	Playing with a ball is suitable	**ball**
him	John, Mary, dog	Mary and dog eliminated	John preferred	Dog can chase John	**John**
they	John, Mary, dog	Requires plural group	Recent group preferred	Group can go home	**John + Mary + dog**
all	John, Mary, dog	Requires plural group	Recent group preferred	Entire group goes home	**John + Mary + dog**

Simple Constraint Graph

Referring Expression

↓

Candidate Antecedents

↓

Gender & Number Agreement

↓

Semantic Compatibility

↓

Recency

↓

Coherence

↓

Final Antecedent

d) Final Resolved Coreference Chains

The final coreference chains are:

1. **John → He → him**
2. **Mary → She**
3. **Ball → it**
4. **John + Mary + dog → they → all**

Rewritten Paragraph with Clear References

John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.

This rewritten paragraph removes the ambiguity caused by pronouns and clearly identifies every entity.

e) Priority Order of Constraints

The constraints can be applied in the following priority order:

1. **Gender and Number Agreement**
2. **Semantic Compatibility**
3. **Coherence**
4. **Recency**

Justification

Gender and number agreement should be applied first because it can immediately eliminate grammatically impossible candidates. For example, **"She" cannot refer to John**.

Semantic compatibility comes next because the selected entity must make sense with the action. For example, Mary can play with a **ball**, while she cannot play with a **park**.

Coherence ensures that the same entity is tracked consistently throughout the paragraph.

Recency is useful as a tie-breaking constraint when multiple candidates remain possible. It generally prefers the most recently mentioned compatible entity.

What Happens If Recency Is Relaxed?

If the **recency constraint is relaxed**, the system can still use gender, number, semantic compatibility, and coherence to resolve references. However, when several candidates are grammatically and semantically possible, the system may have difficulty deciding which antecedent is most appropriate.

For example, if several male persons were mentioned, **“him”** could refer to more than one person. Without recency, the system would need to depend more heavily on semantic compatibility and discourse coherence.

Conclusion

The constraint-based approach successfully resolves the references by combining **gender and number agreement, recency, semantic compatibility, and coherence**. The final interpretation is:

He → John, She → Mary, it → ball, him → John, they/all → John + Mary + dog.

2. **Conversation History:** User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse “exam”, “concentrate”, “you”).
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

ANSWER:

Given Conversation

User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage

a) Relevant Discourse Planning Steps

Before generating the response, the following steps should be performed:

1. **Identify the user's problem** – The user has an important exam tomorrow and is unable to concentrate.
2. **Identify the communicative goal** – Provide useful advice and motivate the student.
3. **Maintain entity coherence** – Continue referring to the **exam**, **concentration**, and **you**.
4. **Select a discourse relation** – Use a **Cause-Effect** relation: difficulty concentrating can affect exam preparation, so a suitable study strategy is suggested.
5. **Provide practical advice** – Suggest taking a short break and then returning to focused study.
6. **Provide encouragement** – Encourage the student to remain confident.
7. **Check constraints** – Ensure the response is 2–3 sentences, contains at least two required keywords, is logical, polite, and positive.

b) Three Possible Responses

Response 1

Since your exam is tomorrow, take a short break first and then return with a clear focus. Study one small topic at a time, and you can feel more confident about your preparation.

Response 2

Because you are finding it difficult to concentrate, take a short break and then study in focused sessions. Stay confident and remind yourself that consistent effort can help you prepare well for the exam.

Response 3

Your exam is tomorrow, so avoid forcing yourself to study continuously; take a short break and come back with better focus. Stay confident and keep studying one topic at a time.

c) Evaluation of the Three Responses

Criteria	**Response 1**	**Response 2**	**Response 3**
Advise + Encourage	✓	✓	✓
Mentions exam	✓	✓	✓
Maintains coherence	✓	✓	✓
Uses “focus”	✓	✓	✓
Uses “break”	✓	✓	✓
Uses “confident”	✓	✓	✓
2 sentences	✓	✓	✓
Positive tone	✓	✓	✓
Logical consistency	✓	✓	✓

Best Response: Response 1

Response 1 is the best choice because it directly addresses the student's concentration problem and provides both practical advice and encouragement. It maintains strong coherence with the original conversation and naturally uses the required concepts **focus, break, and confident**.

It also follows the **Cause-Effect** relation: because the exam is tomorrow, the student should take a short break and then return to focused preparation.

d) Effect of Violating Any Two Constraints

1. Viating Entity Coherence

If entity coherence is violated, the response may suddenly discuss unrelated topics instead of the **exam** and **concentration** problem. This would make the response irrelevant and confusing.

Example:

"You should exercise regularly and buy some new books. The weather is also good today."

This does not properly address the student's immediate concern about the exam.

2. Violating Response Length

If the response-length constraint is violated and the answer becomes very long, it may fail the required 2–3 sentence format. It can also make the advice unnecessarily complicated for a student who is already struggling to concentrate.

3. **Source Sentence:** "The bank by the river flooded after the storm, but it was saved by quick action."

Note: "bank" is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word "bank" and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence.

ANSWER:

Source Sentence

“The bank by the river flooded after the storm, but it was saved by quick action.”

The word **“bank”** is ambiguous. It can mean either a **financial institution** or the **side of a river**.

a) Resolve the Ambiguous Word “Bank” and Justify Using Constraints

The correct meaning of **“bank”** is **riverbank**.

Justification using constraints:

1. **Contextual constraint:**
   The phrase **“by the river”** indicates that the bank is associated with a river.
2. **Semantic compatibility:**
   A riverbank can **flood after a storm**, whereas a financial institution is not the natural interpretation of something that floods in this context.
3. **Discourse coherence:**
   The second clause says **“it was saved by quick action.”** Here, “it” refers to the same riverbank mentioned in the first clause.
4. **Information preservation:**
   Interpreting “bank” as riverbank preserves the complete meaning of the original sentence.

Therefore:

bank → riverbank

b) Predicate Logic Representation

Let the following predicates be used:

- **River(r)** – r is a river.
- **Bank(b)** – b is a bank.
- **Location(b,r)** – b is located by/near r.
- **Storm(s)** – s is a storm.
- **Flood(b)** – b flooded.
- **After(x,y) –** event x occurred after event y.
- **QuickAction(a)** – a is a quick action.

- **Saved(b)** – b was saved.
- **By(b,a)** – b was saved by action a.

Predicate Logic

River(r) ∧ Bank(b) ∧ Location(b,r) ∧

Storm(s) ∧ Flood(b) ∧ After(Flood(b),s) ∧

QuickAction(a) ∧ Saved(b) ∧ By(b,a)

This representation preserves the important entities and relationships: **river, bank, storm, flooding, quick action, and saving**.

c) Target Language Sentence

Simple English Paraphrase

The riverbank flooded after the storm, but quick action saved it from further damage.

Tamil Translation

புயலுக்குப் பிறகு ஆற்றங்கரை வெள்ளத்தில் மூழ்கியது, ஆனால் விரைவான நடவடிக்கையால் அது மேலும் சேதமடைவதிலிருந்து காப்பாற்றப்பட்டது.

The target sentence preserves:

- **Riverbank**
- **Storm**
- **Flooding**
- **Quick action**
- **Saving**
- The **contrast** between the two clauses

d) Simple Discourse Tree – RST Style

The two clauses have a **Contrast** relationship.

WHOLE SENTENCE

|

CONTRAST

/ \

/ \

Clause 1 Clause 2

| |

Bank flooded Quick action

after storm saved it

Explanation

- **Clause 1:** The riverbank flooded after the storm.
- **Clause 2:** Quick action saved the riverbank.
- **Relation: Contrast**

The contrast shows that although the riverbank experienced flooding, quick action produced a positive outcome.

e) Advantages of Constraint-Based Approach over Pure Statistical Machine Translation

A constraint-based approach is useful because it explicitly checks the linguistic and semantic constraints before producing the final interpretation.

Advantages:

1. **Correct ambiguity resolution**
   It correctly identifies **"bank" as riverbank** using the context "by the river."
2. **Semantic consistency**
   It ensures that the selected meaning is compatible with the action **"flooded."**
3. **Preservation of important information**
   Important entities such as the river, bank, storm, flooding, and quick action are retained.
4. **Maintains discourse structure**
   The **Contrast** relationship between flooding and being saved is preserved.

5. **Reduces translation errors**

   Explicit constraints can prevent the system from incorrectly interpreting “bank” as a financial institution.

6. **Better explainability**

   The system can explain why **riverbank** was selected instead of financial bank.

7. **Improved coherence**

   The pronoun **“it”** remains connected to the previously identified riverbank.

Conclusion

The constraint-based approach is more reliable for ambiguous sentences because it combines **contextual, semantic, grammatical, and discourse constraints**. In this sentence, it correctly resolves **“bank” as riverbank**, preserves the meaning of the source sentence, and maintains the **Contrast** relationship between the two clauses.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________